

WORKING WITH FREQUENCY TABLES

Data that deals with averages is often presented in what is known as a frequency table. Frequency tables show the frequency with which certain values appear in a set of data.

Finding the median using a frequency table

The process for finding the median (middle) value from a frequency table depends on whether the total frequency is an odd or an even number.

The following marks were scored in a test and entered in a frequency table:

20, 20, 18, 20, 18, 19, 20, 20

Mark	Frequency	number of times each mark appears
18	2	frequency contains 4th value
19	1 (2 + 1 = 3)	frequency contains 5th value
20	6 (3 + 6 = 9)	median frequency (entry contains 5th value in list)
	9	total frequency

median mark

Because the total frequency of 9 is odd, to find the median first add 1 to it and then divide it by 2. This makes 5, meaning that the 5th value is the median. Count down the frequency column adding the values until reaching the row containing the 5th value. The median mark is 20.

The following marks were scored in a test and entered in a frequency table:

18, 17, 20, 19, 19, 18, 19, 18

Mark	Frequency
17	1
18	3 (1 + 3 = 4)
19	3 (4 + 3 = 7)
20	1 (7 + 1 = 8)
	8

total frequency

The total frequency of 8 is even (8), so there are two middle values (4th and 5th). Count down the frequency column adding values to find them.

▽ **An even total frequency**
If the total frequency is even, the median is calculated from the two middle values.

$$\text{Median} = \frac{\text{1st middle value} + \text{2nd middle value}}{2}$$

$$\frac{18 + 19}{2} = 18.5$$

1st middle value 2nd middle value
median

The two middle values (4th and 5th) represent the marks 18 and 19 respectively. The median is the mean of these two marks, so add them together and divide by 2. The median mark is 18.5.

Finding the mean from a frequency table

To find the mean from a frequency table, calculate the total of all the data as well as the total frequency. Here, the following marks were scored in a test and entered into a table:

16, 18, 20, 19, 17, 19, 18, 17, 18, 19, 16, 19

Mark	Frequency	Mark	Frequency	Total marks (mark × frequency)
16	2	16	2	16 × 2 = 32
17	2	17	2	17 × 2 = 34
18	3	18	3	18 × 3 = 54
19	4	19	4	19 × 4 = 76
20	1	20	1	20 × 1 = 20
			12	216

range of values frequency shows number of times each mark was scored add frequencies together to get total frequency total marks

Enter the given data into a frequency table.

Find the total marks scored by multiplying each mark by its frequency. The total sum of each part of the data is the sum of values.

$$\text{Mean} = \frac{\text{Sum of values}}{\text{Number of values}}$$

total marks total frequency

$$216 \div 12 = 18$$

total marks total frequency mean mark

To find the mean, divide the sum of values, in this example, the total marks, by the number of values, which is the total frequency.